

# JavaScript DOM Assignment #4 – classList Manipulation

## Scenario

In real websites, developers **add, remove, and toggle CSS classes** using JavaScript to:

- Show / hide elements
- Change colors
- Highlight buttons
- Enable dark mode

This is done using `classList`.

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## Task 1: Add a Class

1. Create a paragraph.
2. Create a CSS class named:
3. highlight

(change text color or background)

4. Create a button:
  5. Add Highlight
  6. When button is clicked:
    - Add highlight class to the paragraph.
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## Task 2: Remove a Class

1. Use the same paragraph.
  2. Create a button:
  3. Remove Highlight
  4. When clicked:
    - Remove highlight class from paragraph.
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## Task 3: Toggle a Class

1. Create a button:
  2. Toggle Highlight
  3. When clicked:
    - Toggle the highlight class on paragraph.
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## Task 4: Check Class Exists

1. When button is clicked:
    - Check if paragraph has highlight class
    - Show result using alert():
    - Class exists

or

Class not exists
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## Task 5: Button Active State (Real World)

### Real-life Example

When user clicks a button, it becomes active (like filters in shopping apps).

1. Create a button:
  2. Buy Now
  3. Create CSS class:
  4. active-btn
  5. On button click:
    - Toggle active-btn class
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## Task 6: Dark Mode (Industry Example)

1. Create a button:
2. Dark Mode
3. Create CSS class:
4. dark-mode

(background black, text white)

5. When clicked:
  - Toggle dark-mode on <body>

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## Required JavaScript Methods

Students **must use**:

- `classList.add()`
- `classList.remove()`
- `classList.toggle()`
- `classList.contains()`

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## Instructions for Students

- Write JavaScript only in `script.js`
- Write CSS only in `style.css`
- Use buttons to trigger actions
- Use `alert()` where required
- Do not use inline JavaScript

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## Submission Instructions (GitHub)

1. Create a **public GitHub repository**  
**Name format:**
2. `YourName_DOM_Assignment-4`
3. Upload your files:
  - `index.html`
  - `script.js`
4. Enable **GitHub Pages**:
  - Go to **Settings** → **Pages**
  - Source: `main`
  - Folder: `/root`
  - Save and copy the **live link**
5. Submit on **Google Classroom**:
  - GitHub Repository Link
  - Live Website Link

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## What You Will Learn

- How CSS classes work with JavaScript

- How websites change styles dynamically
- Real-world UI behavior (dark mode, active buttons)
- Clean separation of HTML, CSS, and JS